Zain Ali Butt

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FULL-STACK SOFTWARE DEVELOPER | APPLIED AI

Full-stack developer who takes client work from requirements to production. Delivered paid systems in 2026, with ongoing support. Strongest in TypeScript, Node and PostgreSQL, and in the integration and deployment work around them.

COMMERCIAL EXPERIENCE

Freelance Software Developer Jun 2026 – Present
Self-Employed *London, UK*

- Shipped a paid shipment-operations platform for a cross-border logistics team on Next.js, TypeScript, Supabase and Vercel. Retained for ongoing support.
- Replaced manual tracking and chasing with a single completion workflow. Client reported roughly 60% higher operational efficiency and near-elimination of routine tracking errors.
- Built and launched a booking-enabled website for a private music tutor. Client reported two-to-threefold growth in engagement, enquiries and bookings.

Invigilator Nov 2024 – Present (seasonal)
Loxford School *London, UK*

- Run regulated examinations for 10–220 candidates, covering access arrangements, incidents and secure materials under strict confidentiality.

PROJECTS

ProofLens | *Media provenance and tamper evidence* Oct 2025 – Apr 2026
github.com/zainalibutt/Proof-Lens · proof-lens.vercel.app

- Built and deployed a provenance system across a React Native app, a 35-endpoint Express API, PostgreSQL/Supabase, AWS S3 and a public web verifier.
- Implemented SHA-256 hashing, device-held Ed25519 signing, independent server re-hashing and RFC 3161 timestamping, verifiable offline with OpenSSL.
- Research prototype, evaluated over 36 checks and seven investigations: 124 credentials anchored, 9.6-second mean capture-to-anchor.

River | *Multiplayer poker with solver-backed bots* Aug 2026 – Present
playriver.uk

- Built and launched an early v1 of a server-authoritative Texas Hold'em platform: a pure TypeScript rules engine, a WebSocket server that owns every chip through an append-only ledger, and commit-reveal shuffles. 1,500+ tests in CI.
- Built the bots: the live one solves each heads-up final betting round with a CFR+ game-theory solver and adapts to each player's habits, playing about even with the Slumbot benchmark. In research, trained PyTorch networks on 200,000+ solved positions so it can also solve the round before; an opponent that knows its strategy wins 42% less from that version than from the live bot.

Melody | *Company research terminal* Apr 2026 – Jun 2026
github.com/zainalibutt/Melody · melody-terminal.vercel.app

- Built a command-first terminal over SEC EDGAR and delayed market data in React and Express, surfacing source, freshness and provider-failure states explicitly.

Lancaster Music Hall | *Venue booking and scheduling system* Jan – Apr 2025

- Led six developers through requirements, delivery recovery, testing and handover, rebuilding the calendar, conflict-validation and analytics features in Java and MySQL.

EDUCATION

City St George's, University of London 2023 – 2026
BSc Computer Science, Upper Second-Class Honours (69.3%) *London, UK*

Final-year project 83%. Selected modules: Theory of Computation 85.8%, Object-Oriented Analysis & Design 84.4%, Algorithms 83%, C++ 82.5%.

TECHNICAL SKILLS

Applied: TypeScript, JavaScript, React, Next.js, React Native, Node.js/Express, WebSockets, PostgreSQL/Supabase, REST APIs, AWS S3, GitHub Actions, Playwright, Vercel, Railway, PyTorch, CFR+ solvers, Blender (Python), AI coding agents (Claude, Codex, open-source models).
Foundational: Python, Java, C++, machine learning, databases, networking, operating systems and system architecture.